

Comprehensive UPI Transaction Analysis using SQL

```
1  -- 1. All transactions
2  SELECT * FROM upi_transaction_data utd ;
3
4  -- 2. Transactions above ₹10,000
5  SELECT * FROM upi_transaction_data utd
6  WHERE utd."Amount_INR" > 10000;
7
8  -- 3. Transactions from HDFC bank
9  SELECT * FROM upi_transaction_data utd
10 WHERE utd."Bank_Name" = 'SBI';
11
12 -- 4. Failed transactions
13 SELECT * FROM upi_transaction_data utd
14 WHERE utd."Status" = 'Failed';
15
16 -- 5. Transactions in July 2026
17 SELECT * FROM upi_transaction_data utd
18 WHERE EXTRACT(MONTH FROM to_date(utd."Transaction_Date", 'DD-MM-YYYY')) = 5
19       AND EXTRACT(YEAR FROM to_date(utd."Transaction_Date", 'DD-MM-YYYY')) = 2026;
20
21 -- 6. Total transaction amount per bank
22 SELECT "Bank_Name", SUM("Amount_INR") AS total_amount
23 FROM upi_transaction_data utd
24 GROUP BY "Bank_Name";
25
26 -- 7. Average transaction per merchant
27 SELECT utd."Merchant_Name", AVG("Amount_INR") AS avg_amount
28 FROM upi_transaction_data utd
29 GROUP BY utd."Merchant_Name";
30
31 -- 8. Count of transactions per payment app
32 SELECT utd."UPI_App", COUNT(*) AS txn_count
33 FROM upi_transaction_data utd
34 GROUP BY utd."UPI_App";
35
36 -- 9. Merchants with more than 100 transactions
37 SELECT utd."Merchant_Name", COUNT(*) AS txn_count
38 FROM upi_transaction_data utd
39 GROUP BY utd."Merchant_Name"
40 HAVING COUNT(*) > 100;
41
42 -- 10. Merchant Category with total transaction value above ₹1,00,000
43 SELECT utd."Merchant_Category", SUM("Amount_INR") AS total_value
44 FROM upi_transaction_data utd
45 GROUP BY utd."Merchant_Category"
46 HAVING SUM("Amount_INR") > 100000;
47
48 -- 11. Top 10 highest transactions
49 SELECT * FROM upi_transaction_data utd
50 ORDER BY utd."Amount_INR" DESC
```

```
51  LIMIT 10;
52
53  -- 12. Lowest 5 transactions
54  SELECT * FROM upi_transaction_data utd
55  ORDER BY utd."Amount_INR" ASC
56  LIMIT 5;
57
58  -- 13. Top 5 merchants by transaction volume
59  SELECT utd."Merchant_Name", SUM("Amount_INR") AS total_sales
60  FROM upi_transaction_data utd
61  GROUP BY utd."Merchant_Name"
62  ORDER BY total_sales DESC
63  LIMIT 5;
64
65  -- 14. Most active Customers
66  SELECT utd."Customer_ID", COUNT(*) AS txn_count
67  FROM upi_transaction_data utd
68  GROUP BY UTD."Customer_ID"
69  ORDER BY txn_count DESC
70  LIMIT 10;
71
72  -- 15. Daily totals
73  SELECT DATE("Transaction_Date") AS txn_day, SUM("Amount_INR") AS Total_Amount
74  FROM upi_transaction_data utd
75  GROUP BY txn_day;
76
77  -- 24. Monthly totals
78  SELECT DATE_TRUNC('month', to_date(utd."Transaction_Date", 'DD-MM-YYYY')) AS txn_month,
79  SUM("Amount_INR") AS total
80  FROM upi_transaction_data utd
81  GROUP BY txn_month;
82
83  -- 17. Count distinct Customers
84  SELECT COUNT(DISTINCT "Customer_ID") AS unique_senders
85  FROM upi_transaction_data utd ;
86
87  -- 18. Count distinct receivers
88  SELECT COUNT(DISTINCT "Merchant_Name") AS unique_receivers
89  FROM upi_transaction_data utd ;
90
91  -- 19. Highest transaction per bank
92  SELECT "Bank_Name", MAX("Amount_INR") AS max_txn
93  FROM upi_transaction_data utd
94  GROUP BY "Bank_Name";
95
96  -- 20. Lowest transaction per merchant category
97  SELECT utd."Merchant_Category", MIN("Amount_INR") AS min_txn
98  FROM upi_transaction_data utd
99  GROUP BY "Merchant_Category";
100
101  -- 21. Transactions per Customer per day
102  SELECT utd."Customer_ID", DATE("Transaction_Date"), COUNT(*) AS txn_count
103  FROM upi_transaction_data utd
104  GROUP BY utd."Customer_ID", DATE("Transaction_Date");
```

```
104
105 -- 22. Average transaction per day
106 SELECT DATE("Transaction_Date"), AVG("Amount_INR") AS avg_txn
107 FROM upi_transaction_data utd
108 GROUP BY DATE("Transaction_Date");
109
110 -- 23. Top 3 senders per bank
111 SELECT "Bank_Name", "Customer_ID", txn_count
112 FROM (
113     SELECT utd."Bank_Name", utd."Customer_ID", COUNT(*) AS txn_count,
114            RANK() OVER (PARTITION BY "Bank_Name" ORDER BY COUNT(*) DESC) AS rnk
115     FROM upi_transaction_data utd
116     GROUP BY utd."Bank_Name", utd."Customer_ID"
117 ) t
118 WHERE rnk <= 3;
119
120 -- 24. Match customers with merchants they paid
121 SELECT t1."Customer_ID", t2."Merchant_Name", t1."Amount_INR", t1."Transaction_Date"
122 FROM upi_transaction_data t1
123 JOIN upi_transaction_data t2
124     ON t1."Merchant_Name" = t2."Merchant_Name"
125 WHERE t1."Customer_ID" <> t2."Customer_ID";
126
127 -- 25. Find customers who paid the same merchant multiple times
128 SELECT t1."Customer_ID",
129        t1."Merchant_Name",
130        t1."Amount_INR",
131        t2."Amount_INR"
132 FROM upi_transaction_data t1
133 JOIN upi_transaction_data t2
134     ON t1."Customer_ID" = t2."Customer_ID"
135     AND t1."Merchant_Name" = t2."Merchant_Name"
136     AND t1."Transaction_ID" <> t2."Transaction_ID";
137
138 -- 26. Customers with their highest transaction
139 SELECT t."Customer_ID",
140        t."Transaction_ID",
141        t."Amount_INR"
142 FROM upi_transaction_data t
143 JOIN (
144     SELECT "Customer_ID", MAX("Amount_INR") AS max_amt
145     FROM upi_transaction_data
146     GROUP BY "Customer_ID"
147 ) sub
148     ON t."Customer_ID" = sub."Customer_ID"
149     AND t."Amount_INR" = sub.max_amt;
150
151 -- 27. Merchants with their top customers
152 SELECT t."Customer_ID",
153        t."Merchant_Name",
154        t."Amount_INR"
155 FROM upi_transaction_data t
156 JOIN (
157     SELECT "Merchant_Name", MAX("Amount_INR") AS max_amt
```

```
158         FROM upi_transaction_data
159         GROUP BY "Merchant_Name"
160     ) sub
161     ON t."Merchant_Name" = sub."Merchant_Name"
162     AND t."Amount_INR" = sub.max_amt;
163
164 -- 28. Rank customers by spending per merchant
165 SELECT t."Customer_ID",
166        t."Merchant_Name",
167        SUM(t."Amount_INR") AS total_spent,
168        RANK() OVER (PARTITION BY t."Merchant_Name" ORDER BY SUM(t."Amount_INR") DESC)
169 AS rank_in_merchant
170 FROM upi_transaction_data t
171 GROUP BY t."Customer_ID", t."Merchant_Name";
172
173 -- 29. Transactions above average amount
174 SELECT * FROM upi_transaction_data
175 WHERE "Amount_INR" > (SELECT AVG("Amount_INR")
176 FROM upi_transaction_data);
177
178 -- 30. Merchants with avg txn > overall avg
179 SELECT "Merchant_Name", AVG("Amount_INR") AS avg_amount
180 FROM upi_transaction_data
181 GROUP BY "Merchant_Name"
182 HAVING AVG("Amount_INR") > (SELECT AVG("Amount_INR")
183 FROM upi_transaction_data);
184
185 -- 31. Customers with more than 50 transactions
186 SELECT "Merchant_Name", AVG("Amount_INR") AS avg_amount
187 FROM upi_transaction_data
188 GROUP BY "Merchant_Name"
189 HAVING AVG("Amount_INR") > (SELECT AVG("Amount_INR")
190 FROM upi_transaction_data);
191
192 -- 32. Transactions from top 5 banks by volume
193 SELECT * FROM upi_transaction_data
194 WHERE "Bank_Name" IN (
195     SELECT "Bank_Name"
196     FROM upi_transaction_data
197     GROUP BY "Bank_Name"
198     ORDER BY SUM("Amount_INR") DESC
199     LIMIT 5
200 );
201
202 -- 33. Rank transactions by amount
203 SELECT "Transaction_ID",
204        "Amount_INR",
205        RANK() OVER (ORDER BY "Amount_INR" DESC) AS rank_amount
206 FROM upi_transaction_data;
207
208 -- 34. Running total per customers
209 SELECT "Transaction_ID",
210        "Amount_INR",
211        RANK() OVER (ORDER BY "Amount_INR" DESC) AS rank_amount
```

```
211 FROM upi_transaction_data;
212
213 -- 35. Average transaction per bank
214 SELECT "Bank_Name",
215        "Transaction_ID",
216        "Amount_INR",
217        AVG("Amount_INR") OVER (PARTITION BY "Bank_Name") AS avg_per_bank
218 FROM upi_transaction_data;
219
220 -- 36. Top transaction per merchant
221 SELECT "Bank_Name",
222        "Transaction_ID",
223        "Amount_INR",
224        AVG("Amount_INR") OVER (PARTITION BY "Bank_Name") AS avg_per_bank
225 FROM upi_transaction_data;
226
227
228
229
230
231
```